

CRD LAB: BY AKSHAT GUPTA

Flipkart  Case study

CRD



Custom Resource



Operator



Deployment



Pods

Lab: Build Your Own FlipkartApplication Operator

Developer



FlipkartApplication



Python Operator (Kopf)



Deployment



Pods

PROBLEM STATEMENT:

In **Flipkart**, developers were repeatedly creating Deployment YAML files for every application, leading to duplication, manual effort, and configuration mistakes. The DevOps team wanted a simpler way for developers to deploy applications without writing complex Kubernetes manifests. To solve this, they introduced a custom resource called **FlipkartApplication**.

Developers now only provide application details such as image and replicas, while an Operator automatically creates the required Kubernetes Deployment and Pods. This standardizes deployments and reduces operational overhead.

```
kubectl get nodes
```

Step 2: Install python

```
sudo apt update
```

```
sudo apt install python3 python3-pip python3-venv -y
```

Verify: python3 --version

Step 3: Create Virtual Environment

```
mkdir /root/operator-lab
```

```
cd /root/operator-lab
```

```
python3 -m venv venv
```

Activate:

```
cd /home/ubuntu
```

```
source /root/operator-lab/venv/bin/activate
```

Verify:

```
which python
```

Expected:

/root/operator-lab/venv/bin/python

Step 4: Install Operator Libraries

pip install kopf kubernetes

Verify:

pip list | grep kopf

Expected: kopf

Step 5: Create CRD

vi flipkart-crd.yaml

Copy the yaml script from <https://github.com/letslearn207/kubernetes/blob/main/c13/flipkart-crd.yaml>

kubectl apply -f flipkart-crd.yaml

Verify:

kubectl get crd

o/p:

flipkartapplications.flipkart.com

Step 6:

Verify New Resource Exists

kubectl api-resources | grep flipkart

Expected:

flipkartapplications

Step 7: Create Operator

```
vi flipkart_operator.py
```

https://github.com/letslearn207/kubernetes/blob/main/c13/flipkart_operator.py

Step 8: Start Operator

Open Terminal 1

```
cd /root/operator-lab
```

```
source venv/bin/activate
```

```
kopf run flipkart_operator.py
```

Expected:

Initial authentication has finished

KEEP TERMINAL OPEN

Step 9: Create Custom Resource

Open Terminal 2

File:

```
vi flipkartapp.yml
```

open and take from

<https://github.com/letslearn207/kubernetes/blob/main/c13/flipkartapp.yml>

Apply:

```
kubectl apply -f flipkartapp.yml
```

Expected:

flipkartapplication.flipkart.com/payment-service created

Step 10: Verify CR

```
kubectl get flipkartapplications
```

Expected:

NAME

payment-service

Step 11: Verify Deployment

kubectrl get deployment

Expected:

NAME

payment-service

Step 12: Verify pod

kubectrl get pods

payment-service-xxxxx

payment-service-yyyyy

payment-service-zzzzz
